Term Project : Predictive Analytics

Predicting Customer Churn in a Telecommunications Company

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Description: A clustering based approach to customer segmentation and churn risk analysis

Introduction and Business Problem

In 2026, telecommunications and SaaS companies operate in highly competitive markets where customer acquisition costs are significantly higher than customer retention costs. As a result, customer churn has become a critical performance metric that directly affects profitability and long-term growth. Many organizations still respond to churn reactively, often identifying dissatisfied customers only after cancellation has already occurred.

This project addresses that challenge by using predictive analytics to identify churn risk before customers leave. The objective is to move from reactive decision making to proactive intervention, enabling businesses to detect early warning patterns in customer behavior and take timely retention actions.

The value of this project extends beyond generating churn probabilities. A key goal is to uncover the underlying drivers of churn, such as contract structure, service reliability issues, and customer experience pain points. By identifying these factors, organizations can design targeted and personalized retention strategies instead of relying on broad, generic incentives. This approach supports more effective customer engagement, improved retention outcomes, and stronger revenue stability.

Problem Statement

The goal of this project is to build a predictive model that identifies which customers are at high risk of leaving a telecom provider. Rather than just providing a simple probability score, I want to uncover the specific drivers of churn, such as contract types or service glitches. By identifying these early red flags, the business can intervene with targeted retention

efforts that actually address the reasons why a customer is unhappy. I will be following the CRISP-DM frameworkc Cross Industry Standard Process for Data Mining to ensure that the technical work stays aligned with these business goals.

Project Objective

This project has three primary objectives:

1. Build a predictive classification model to distinguish customers likely to churn from those who will stay.
2. Identify the key features and customer attributes that are most associated with churn behavior.
3. Provide actionable business recommendations that allow retention teams to intervene with high-risk customer segments before churn occurs.

Research Questions

1. Which customer attributes (contract type, tenure, service usage, payment method) are most strongly associated with churn?
2. How accurately can customer churn be predicted using available demographic and account data?
3. Which classification model performs best for this dataset based on Recall, F1 score, and ROC-AUC metrics that reflect the business cost of missing a churning customer?

Dataset Description

This project uses a telecom customer churn dataset containing demographic attributes, service subscription details, account information, and customer outcomes. The target variable is churn status (Yes/No). Features include contract type, tenure, monthly charges, total charges, internet services, payment method, and other service related indicators.

First thing first, import core libraries required.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import GridSearchCV
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score,roc_curve, auc
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.ensemble import (
    RandomForestClassifier, ExtraTreesClassifier,AdaBoostClassifier,
    GradientBoostingClassifier)
from sklearn.dummy import DummyClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    roc_auc_score, confusion_matrix, classification_report
)

import warnings
warnings.filterwarnings("ignore")
```

Data Loading and Exploration

I start by loading the training and test data from HuggingFace. I keep them separate from the beginning to avoid any data leakage. The training set is what I use to build and learn patterns, and the test set is locked away until the very end when I evaluate my final model.

In [2]:
```
# Load data using hugging face provided pandas read_csv
base = "hf://datasets/aai510-group1/telco-customer-churn/"

churn_train = pd.read_csv(base + "train.csv")
churn_test  = pd.read_csv(base + "test.csv")

print("Train shape:", churn_train.shape)
print("Test shape:", churn_test.shape)
```

```
Train shape: (4225, 52)
Test shape: (1409, 52)
```

Next, I look at the target variable to understand the class balance. This helps me see if I have a churn problem upfront. If one class dominates, I need to handle it carefully during modeling to avoid a model that just predicts the majority class and ignores the minority.

In [3]:
```
print("Train shape:", churn_train.shape)
print("\nTarget distribution:")
print(churn_train["Churn"].value_counts(dropna=False))
print("\nTarget proportion:")
print(churn_train["Churn"].value_counts(normalize=True).round(4))
```

```
Train shape: (4225, 52)

Target distribution:
Churn
0    3104
1    1121
Name: count, dtype: int64

Target proportion:
Churn
0    0.7347
1    0.2653
Name: proportion, dtype: float64
```

I have 4,225 training records, and my target is imbalanced with about 73.5% non-churn and 26.5% churn customers. This tells me churn is the minority class, so I cannot rely on accuracy alone.

I check for missing values because they can break my model during training. Some columns might be incomplete, and I need to decide whether to drop them, fill them, or handle them in some other way before I can proceed.

In [4]:
```
print("\nTop missing columns (train):")
print(churn_train.isnull().sum().sort_values(ascending=False).head(15))
```

```
Top missing columns (train):
Churn Category          3104
Churn Reason            3104
Offer                   2324
Internet Type            886
Age                        0
Streaming Movies           0
Payment Method             0
Phone Service              0
Population                 0
Premium Tech Support       0
Quarter                    0
Referred a Friend          0
Satisfaction Score         0
Senior Citizen             0
State                      0
dtype: int64
```

I can see that most missing values are concentrated in Churn Category and Churn Reason, which is expected because those fields are only populated for customers who churned. I also see missing values in Offer and Internet Type, and I treat those as valid business states (no offer / no internet service) rather than random data errors. Since the rest of the columns show zero missing values, my data quality is strong after handling these key fields.

I explore a few specific columns to understand what values they contain. This helps me see if a column is categorical or numeric, and whether there are patterns like missing categories that need filling.

In [5]:
```
# Distinct values for the two columns
for col in ["Offer", "Internet Type", "Churn"]:
    vals = sorted(churn_train[col].dropna().astype(str).unique().tolist())
    print(f"\n{col} | distinct count = {len(vals)}")
    print(vals)
```

```
Offer | distinct count = 5
['Offer A', 'Offer B', 'Offer C', 'Offer D', 'Offer E']

Internet Type | distinct count = 3
['Cable', 'DSL', 'Fiber Optic']

Churn | distinct count = 2
['0', '1']
```

Data Cleaning

Now I remove columns that would leak information about the outcome. Churn Reason and Churn Category only exist after a customer churns, so they cannot be used to predict churn in a real world scenario. Similarly, Churn Score, Customer Status, and Satisfaction Score are computed after churn happens. I drop these from both train and test to ensure clean, realistic predictions.

In [6]:
```
# drop post churn outcome descriptor columns churn reason and churn category
churn_train.drop(columns=['Churn Reason', 'Churn Category','Churn Score',
                          'Customer Status', 'Satisfaction Score'], inplace=True)

churn_test.drop(columns=['Churn Reason', 'Churn Category','Churn Score',
                         'Customer Status', 'Satisfaction Score'], inplace=True)

churn_train.head(4)
```

Out[6]:

	Age	Avg Monthly GB Download	Avg Monthly Long Distance Charges	City	CLTV	Contract	Country	Customer ID	Dependents	Device Protection Plan	...	Tenure in Months	Total Charges
0	72	4	19.44	San Mateo	4849	Two Year	United States	4526-ZJJTM	0	1	...	25	2191.15
1	27	59	45.62	Sutter Creek	3715	Month-to-Month	United States	5302-BDJNT	0	1	...	35	3418.20
2	59	0	16.07	Santa Cruz	5092	Month-to-Month	United States	5468-BPMMO	0	0	...	46	851.20
3	25	27	0.00	Brea	2068	One Year	United States	2212-LYASK	0	1	...	27	1246.40

4 rows × 47 columns

For Offer and Internet Type, the missing values represent a real business state, not gaps in data collection. A customer with no offer listed likely truly has no offer. A missing Internet Type means no internet service. I fill these with business-meaningful labels so the model can learn from these states.

In [7]:
```
# Fill missing only for Offer and Internet Type (train + test, consistent rules)
churn_train["Offer"] = churn_train["Offer"].fillna("No Offer")
churn_test["Offer"] = churn_test["Offer"].fillna("No Offer")

churn_train["Internet Type"] = churn_train["Internet Type"].fillna("No Internet Service")
churn_test["Internet Type"] = churn_test["Internet Type"].fillna("No Internet Service")
```

In [8]:
```
print("Top missing columns (train):")
print(churn_train.isnull().sum().sort_values(ascending=False).head(15))
```

```
Top missing columns (train):
Age                      0
Streaming Music          0
Payment Method           0
Phone Service            0
Population               0
Premium Tech Support     0
Quarter                  0
Referred a Friend        0
Senior Citizen           0
State                    0
Streaming Movies         0
Streaming TV             0
Paperless Billing        0
Tenure in Months         0
Total Charges            0
dtype: int64
```

Exploratory Data Analysis

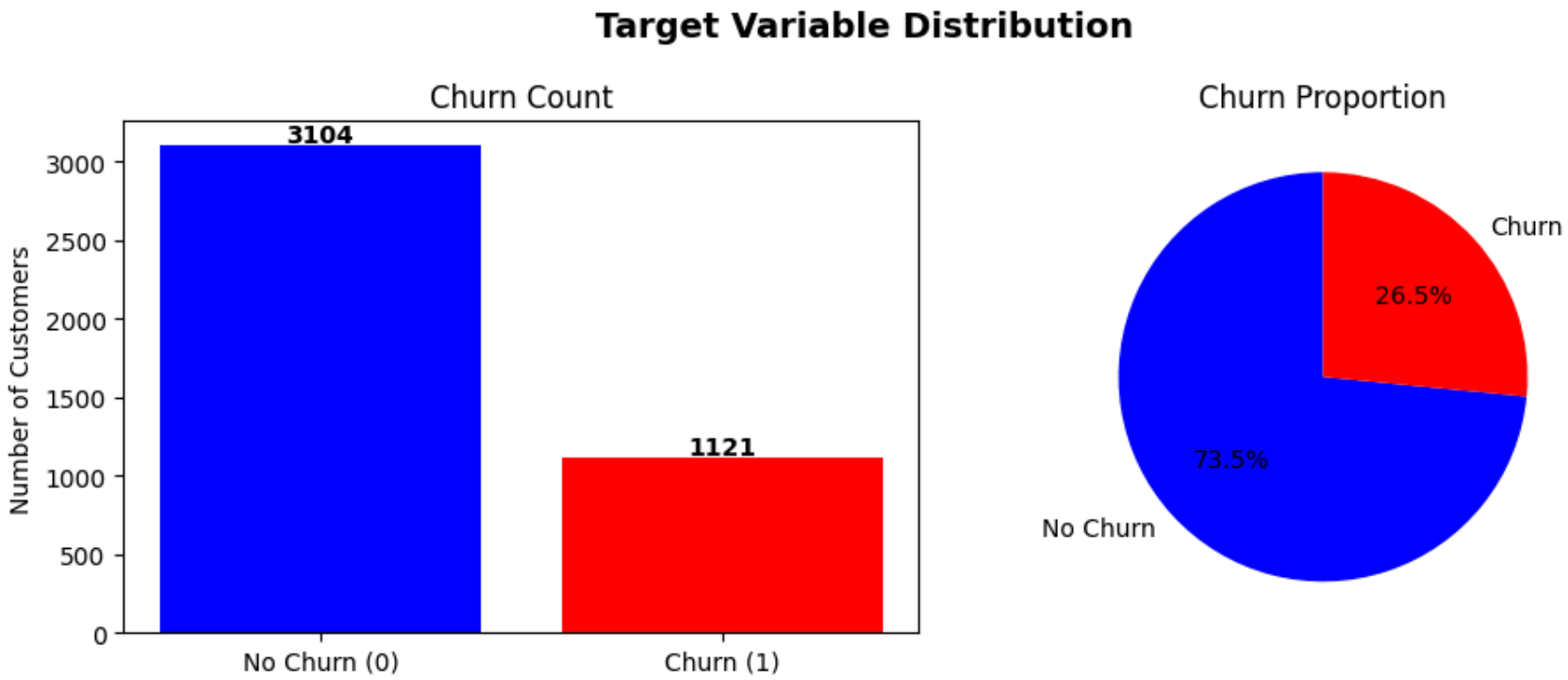
Before modeling, I explore the training data to understand the distribution of key variables, identify patterns associated with churn, and form hypotheses about which features are likely to be predictive. All EDA is performed on the training set only to avoid any influence from the test set.

In the below chart I wanted to label each bar with its exact count so the chart is easier to read.

I was struggling and learned this today that with enumerate(churn_counts.values), I get both the bar position (i) and the bar height (v) at the same time. Then I place text using axes[0].text(i, v + 20, str(v), ha="center", fontweight="bold"), which writes the value slightly above each bar, centered and bold.

In [9]:
```python
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

# Count
churn_counts = churn_train["Churn"].value_counts()
axes[0].bar(["No Churn (0)", "Churn (1)"], churn_counts.values, color=["blue", "red"])
axes[0].set_title("Churn Count")
axes[0].set_ylabel("Number of Customers")
for i, v in enumerate(churn_counts.values):
    axes[0].text(i, v + 20, str(v), ha="center", fontweight="bold")
```

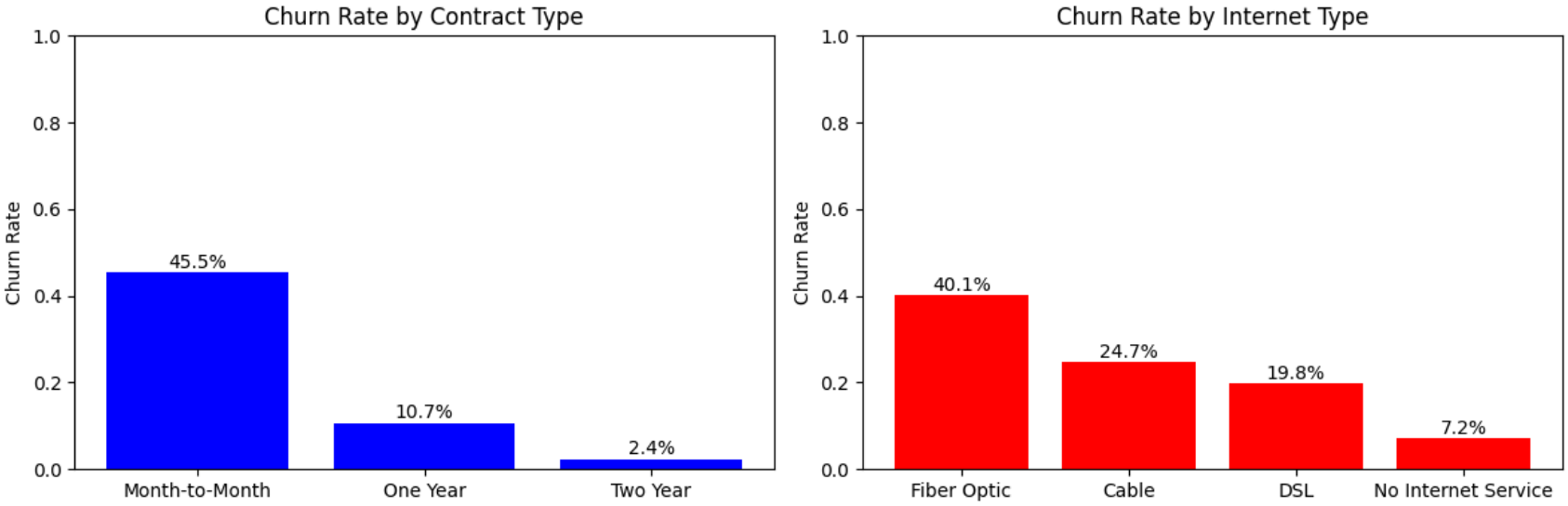
```
# Proportion
axes[1].pie(churn_counts.values, labels=["No Churn", "Churn"],
            autopct="%1.1f%%", colors=["blue", "red"], startangle=90)
axes[1].set_title("Churn Proportion")

plt.suptitle("Target Variable Distribution", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.show()
```



This chart shows my target is imbalanced as the results already confirmed a few steps back: about 73.5% customers did not churn (3104) and 26.5% did churn (1121). I use this to confirm that a model could get decent accuracy by predicting mostly non-churn, so accuracy alone can be misleading. That is why I focus on recall, F1, and ROC-AUC for churn prediction.

In [10]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Contract Type
ct = churn_train.groupby("Contract")["Churn"].mean().sort_values(ascending=False)
```

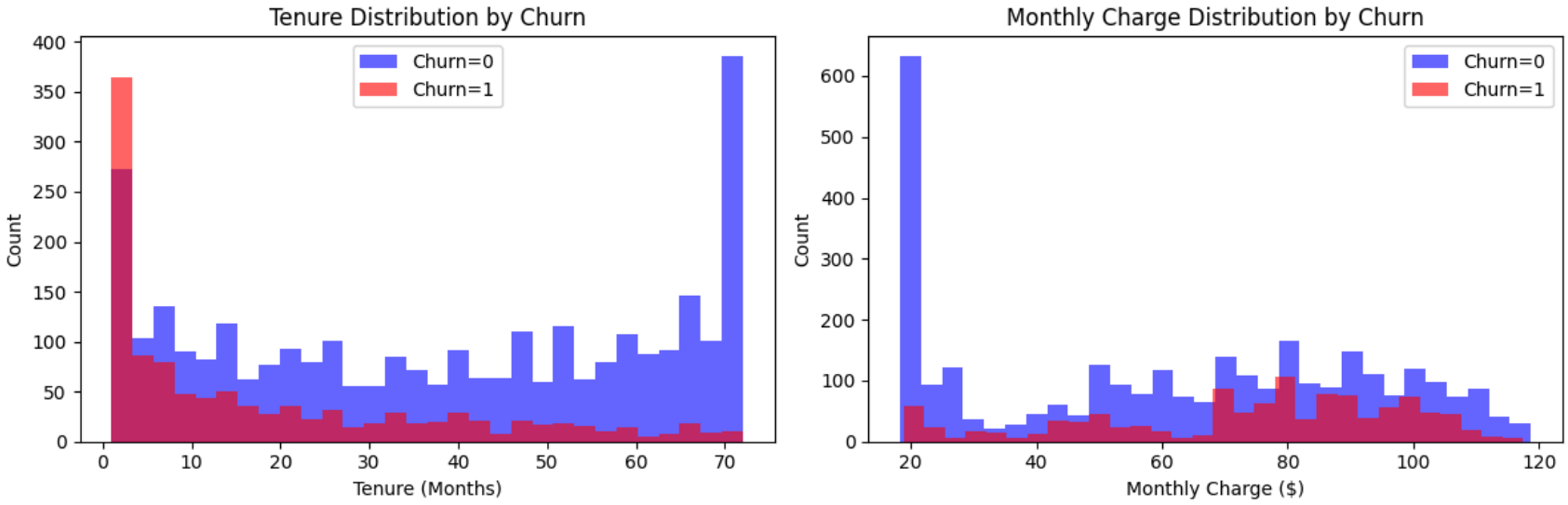
```
axes[0].bar(ct.index, ct.values, color="blue")
axes[0].set_title("Churn Rate by Contract Type")
axes[0].set_ylabel("Churn Rate")
axes[0].set_ylim(0, 1)
for i, v in enumerate(ct.values):
    axes[0].text(i, v + 0.01, f"{v:.1%}", ha="center")

# Internet Type
it = churn_train.groupby("Internet Type")["Churn"].mean().sort_values(ascending=False)
axes[1].bar(it.index, it.values, color="red")
axes[1].set_title("Churn Rate by Internet Type")
axes[1].set_ylabel("Churn Rate")
axes[1].set_ylim(0, 1)
for i, v in enumerate(it.values):
    axes[1].text(i, v + 0.01, f"{v:.1%}", ha="center")

plt.tight_layout()
plt.show()
```



I can see that contract length is a strong churn driver: Month to Month customers churn the most (45.5%), while One Year (10.7%) and Two Year (2.4%) customers are much more stable. I also see churn is highest for Fiber Optic users (40.1%), followed by Cable (24.7%) and DSL (19.8%), and lowest for customers with no internet service (7.2%). This tells me I should focus retention efforts on Month to Month and Fiber Optic segments first.

In [11]:

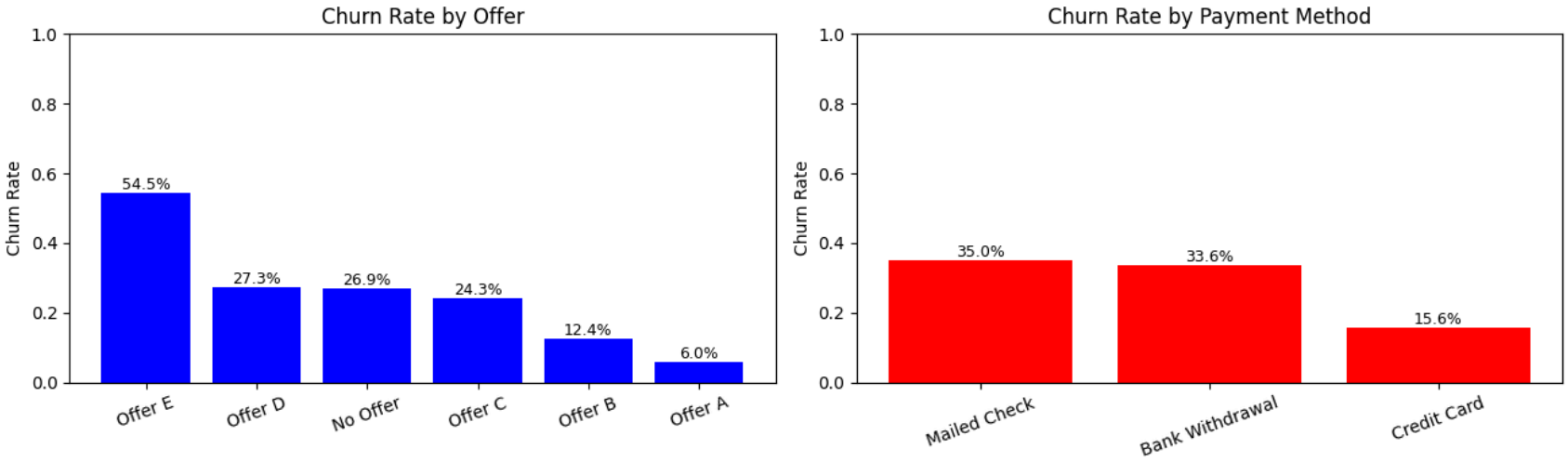
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Tenure distribution by churn
for label, color in zip([0, 1], ["blue", "red"]):
    subset = churn_train[churn_train["Churn"] == label]["Tenure in Months"]
    axes[0].hist(subset, bins=30, alpha=0.6, color=color, label=f"Churn={label}")
axes[0].set_title("Tenure Distribution by Churn")
axes[0].set_xlabel("Tenure (Months)")
axes[0].set_ylabel("Count")
axes[0].legend()

# Monthly charges by churn
for label, color in zip([0, 1], ["blue", "red"]):
    subset = churn_train[churn_train["Churn"] == label]["Monthly Charge"]
    axes[1].hist(subset, bins=30, alpha=0.6, color=color, label=f"Churn={label}")
axes[1].set_title("Monthly Charge Distribution by Churn")
axes[1].set_xlabel("Monthly Charge ($)")
axes[1].set_ylabel("Count")
axes[1].legend()

plt.tight_layout()
plt.show()
```



This plot shows that customers who churn (Churn=1) are concentrated at lower tenure, while non-churn customers (Churn=0) are much more common at long tenure (especially near the upper end). For monthly charges, churners appear more concentrated in higher charge ranges, while non churners include a large low charge group plus a wider spread overall. I interpret this as early life, higher bill customers being the highest risk segment for churn.

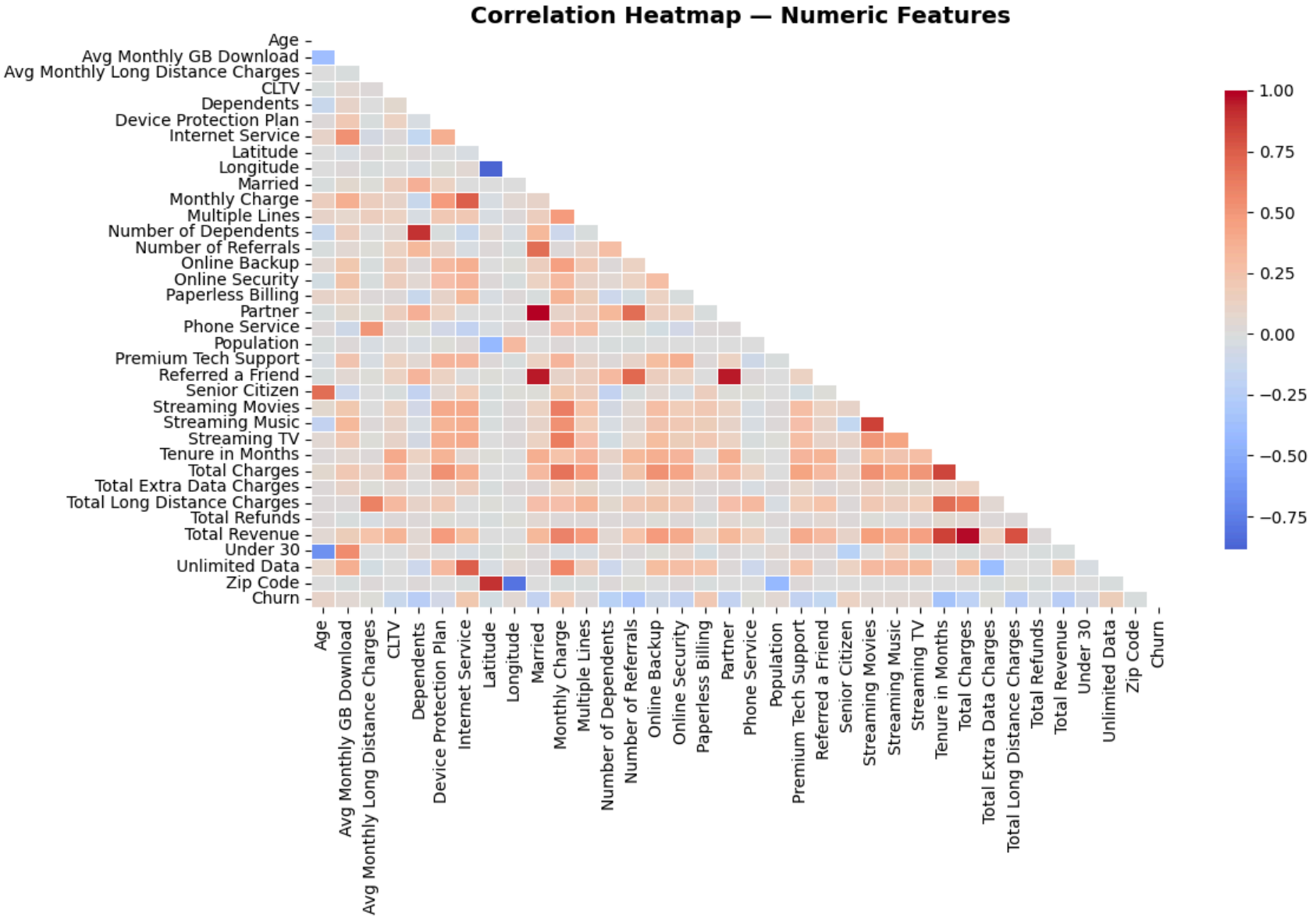
In this code below, I first group customers by Payment Method and take the mean of Churn, so each bar becomes a churn rate for that payment type. I sort those rates from highest to lowest, draw the bars, and keep the y-axis from 0 to 1 because these are proportions, not raw counts. Then I rotate the x labels for readability and use enumerate(pm.values) to place a percentage label above each bar; I learned this today, and it makes charts much easier to interpret quickly.

In [12]:
```
fig, axes = plt.subplots(1, 2, figsize=(13, 4))

# Offer
off = churn_train.groupby("Offer")["Churn"].mean().sort_values(ascending=False)
axes[0].bar(off.index, off.values, color="blue")
axes[0].set_title("Churn Rate by Offer")
axes[0].set_ylabel("Churn Rate")
axes[0].set_ylim(0, 1)
axes[0].tick_params(axis="x", rotation=20)
for i, v in enumerate(off.values):
    axes[0].text(i, v + 0.01, f"{v:.1%}", ha="center", fontsize=9)

# Payment Method
pm = churn_train.groupby("Payment Method")["Churn"].mean().sort_values(ascending=False)
axes[1].bar(pm.index, pm.values, color="red")
axes[1].set_title("Churn Rate by Payment Method")
axes[1].set_ylabel("Churn Rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=20)
for i, v in enumerate(pm.values):
    axes[1].text(i, v + 0.01, f"{v:.1%}", ha="center", fontsize=9)

plt.tight_layout()
plt.show()
```



This graph shows me that churn varies a lot by offer type and payment method. I see the highest churn in Offer E (54.5%), and much lower churn in Offer A/B, which suggests some offers are strongly linked to retention risk. For payment methods, Mailed Check and Bank Withdrawal have much higher churn than Credit Card, so I would prioritize those groups for retention actions.

In [13]:
```
numeric_cols = churn_train.select_dtypes(include=[np.number]).columns.tolist()
corr = churn_train[numeric_cols].corr()

plt.figure(figsize=(12, 8))
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, mask=mask, annot=False, fmt=".2f", cmap="coolwarm",
            center=0, linewidths=0.5, cbar_kws={"shrink": 0.8})
plt.title("Correlation Heatmap — Numeric Features", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.show()
```



This heatmap shows how strongly my numeric features move together: red means positive correlation, blue means negative, and light colors mean weak relationships. I can see strong positive blocks among usage/revenue variables (like monthly charges, total charges, and total revenue), which suggests some features carry overlapping information. For churn, the colors

look mostly moderate rather than extreme, so no single numeric feature alone explains churn; the model needs combined patterns across features.

Feature Engineering

Now I separate my features from my target. The target is the Churn column (0 or 1), and everything else becomes my feature set X. I do this for both train and test to keep them aligned.

In [14]:
```
# Define features/target

y_train = churn_train["Churn"].astype(int)
y_test  = churn_test["Churn"].astype(int)
X_train = churn_train.drop(columns=["Churn"])
X_test = churn_test.drop(columns=["Churn"])
```

My model needs numbers to learn from, but many of my features are categorical text (like contract type or internet type). I use onehot encoding to convert these categories into binary columns. I also align the train and test sets so they have the exact same features in the exact same order.

In [15]:
```
# One-hot encode and align train/test columns
X_train = pd.get_dummies(X_train, drop_first=True)
X_test = pd.get_dummies(X_test, drop_first=True)
X_train, X_test = X_train.align(X_test, join="left", axis=1, fill_value=0)
```

Baseline Model

I start with a Dummy Classifier that always predicts the most common class. This is not a real prediction model, but it gives me a baseline to compare against. If my fancy models cannot beat this simple baseline, they are not working.

In [16]:
```
dummy = DummyClassifier(strategy="most_frequent")
dummy.fit(X_train, y_train)
```

Out[16]:

▾ DummyClassifier ⓘ ⓘ

▸ Parameters

In [17]:
```
dummy_pred = dummy.predict(X_test)
dummy_prob = dummy.predict_proba(X_test)[:, 1]
```

In [18]:
```
print("Dummy Classifier (most_frequent)")
print("Accuracy :", round(accuracy_score(y_test, dummy_pred), 4))
print("Precision:", round(precision_score(y_test, dummy_pred, zero_division=0), 4))
print("Recall   :", round(recall_score(y_test, dummy_pred, zero_division=0), 4))
print("F1       :", round(f1_score(y_test, dummy_pred, zero_division=0), 4))
print("ROC_AUC  :", round(roc_auc_score(y_test, dummy_prob), 4))
print("Confusion Matrix:\n", confusion_matrix(y_test, dummy_pred))
print("classification_report: \n",classification_report(y_test, dummy_pred, zero_division=0))
```

```
Dummy Classifier (most_frequent)
Accuracy : 0.7346
Precision: 0.0
Recall   : 0.0
F1       : 0.0
ROC_AUC  : 0.5
Confusion Matrix:
 [[1035    0]
 [ 374    0]]
classification_report:
               precision    recall  f1-score   support

           0       0.73      1.00      0.85      1035
           1       0.00      0.00      0.00       374

    accuracy                           0.73      1409
   macro avg       0.37      0.50      0.42      1409
weighted avg       0.54      0.73      0.62      1409
```

To establish a performance floor, I first trained a Dummy Classifier using the most frequent class strategy. This model predicted only the non-churn class and produced:

Accuracy: 0.7346
Precision (churn class): 0.0000
Recall (churn class): 0.0000
F1-score (churn class): 0.0000
ROC AUC: 0.5000

These results confirm that class imbalance is present and that accuracy alone is not a reliable metric for churn prediction.

I then trained a Logistic Regression baseline model for comparison. The objective is to verify that the predictive model performs meaningfully better than the dummy benchmark, especially on churn-class Recall, F1-score, and ROC AUC.

A good churn model must outperform the dummy baseline, particularly in identifying churn customers (class 1), not just in overall accuracy.

Model Building

Now I train my first real model, a simple Logistic Regression. This is a linear model that learns a decision boundary between churn and non-churn. It serves as a quick sanity check and a lightweight baseline.

In [19]:
```
# Train and evaluate
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

Out[19]:

▾ LogisticRegression ⓘ ?

▸ Parameters

In [20]:
```
pred = model.predict(X_test)
prob = model.predict_proba(X_test)[:, 1]
```

In [21]:
```
print("Baseline Logistic Regression (Test)")
print("Accuracy :", round(accuracy_score(y_test, pred), 4))
print("Precision:", round(precision_score(y_test, pred, zero_division=0), 4))
print("Recall   :", round(recall_score(y_test, pred, zero_division=0), 4))
print("F1       :", round(f1_score(y_test, pred, zero_division=0), 4))
print("ROC_AUC  :", round(roc_auc_score(y_test, prob), 4))
print("Confusion Matrix:\n", confusion_matrix(y_test, pred))
print("classification_report: \n",classification_report(y_test, pred, zero_division=0))
```

```
Baseline Logistic Regression (Test)
Accuracy : 0.7977
Precision: 0.644
Recall   : 0.5321
F1       : 0.5827
ROC_AUC  : 0.8341
Confusion Matrix:
 [[925 110]
 [175 199]]
classification_report:
              precision    recall  f1-score   support

           0       0.84      0.89      0.87      1035
           1       0.64      0.53      0.58       374

    accuracy                           0.80      1409
   macro avg       0.74      0.71      0.72      1409
weighted avg       0.79      0.80      0.79      1409
```

The logistic regression I just trained does not account for class imbalance. To give the minority churn class more weight, I retrain it with class_weight set to balanced. This tells the model to penalize errors on churn customers more heavily.

In [22]:
```
lr_bal = LogisticRegression(max_iter=1000, class_weight="balanced")
lr_bal.fit(X_train, y_train)

pred_lr_bal = lr_bal.predict(X_test)
prob_lr_bal = lr_bal.predict_proba(X_test)[:, 1]

print("Logistic Regression (class_weight=balanced)")
print("Accuracy :", round(accuracy_score(y_test, pred_lr_bal), 4))
print("Precision:", round(precision_score(y_test, pred_lr_bal, zero_division=0), 4))
print("Recall   :", round(recall_score(y_test, pred_lr_bal, zero_division=0), 4))
print("F1       :", round(f1_score(y_test, pred_lr_bal, zero_division=0), 4))
print("ROC_AUC  :", round(roc_auc_score(y_test, prob_lr_bal), 4))
print("Confusion Matrix:\n", confusion_matrix(y_test, pred_lr_bal))
print("Classification Report:\n", classification_report(y_test, pred_lr_bal, zero_division=0))
```

```
Logistic Regression (class_weight=balanced)
Accuracy : 0.7289
Precision: 0.4935
Recall   : 0.8182
F1       : 0.6157
ROC_AUC  : 0.8335
Confusion Matrix:
 [[721 314]
 [ 68 306]]
Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.70      0.79      1035
           1       0.49      0.82      0.62       374

    accuracy                           0.73      1409
   macro avg       0.70      0.76      0.70      1409
weighted avg       0.80      0.73      0.74      1409
```

Random Forest is more powerful than logistic regression because it can capture nonlinear patterns. But it has many hyperparameters I can tune. Instead of guessing, I use GridSearchCV to automatically test different combinations. I train on 5 folds of the training set and pick the parameters that give the best average performance. Then I evaluate that best model on the test set.

In [23]:
```python
rf_base = RandomForestClassifier(class_weight="balanced", random_state=42)

rf_param_grid = {
    "n_estimators": [200, 400],
    "max_depth": [None, 8, 12],
    "min_samples_split": [2, 10],
    "min_samples_leaf": [1, 5]
}

rf_grid = GridSearchCV(
    estimator=rf_base,
    param_grid=rf_param_grid,
    scoring="roc_auc",
    cv=5,
    n_jobs=-1,
    verbose=1
```

```
)

rf_grid.fit(X_train, y_train)

print("Best RF params:", rf_grid.best_params_)
print("Best CV ROC-AUC:", round(rf_grid.best_score_, 4))
```

```
Fitting 5 folds for each of 24 candidates, totalling 120 fits
Best RF params: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 400}
Best CV ROC-AUC: 0.8831
```

In [24]:

```
best_rf = rf_grid.best_estimator_

pred_rf = best_rf.predict(X_test)
prob_rf = best_rf.predict_proba(X_test)[:, 1]

print("Random Forest (class_weight=balanced)")
print("Accuracy :", round(accuracy_score(y_test, pred_rf), 4))
print("Precision:", round(precision_score(y_test, pred_rf, zero_division=0), 4))
print("Recall   :", round(recall_score(y_test, pred_rf, zero_division=0), 4))
print("F1       :", round(f1_score(y_test, pred_rf, zero_division=0), 4))
print("ROC_AUC  :", round(roc_auc_score(y_test, prob_rf), 4))
print("Confusion Matrix:\n", confusion_matrix(y_test, pred_rf))
print("Classification Report:\n", classification_report(y_test, pred_rf, zero_division=0))
```

```
Random Forest (class_weight=balanced)
Accuracy : 0.8254
Precision: 0.7319
Recall   : 0.5401
F1       : 0.6215
ROC_AUC  : 0.8924
Confusion Matrix:
 [[961  74]
 [172 202]]
Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.93      0.89      1035
           1       0.73      0.54      0.62       374

    accuracy                           0.83      1409
   macro avg       0.79      0.73      0.75      1409
weighted avg       0.82      0.83      0.82      1409
```

Extra Trees is similar to Random Forest but with randomized thresholds at each split. This makes it faster and sometimes more generalizable. I train it with the same class_weight balance to handle the imbalanced classes.

In [25]:
```python
et = ExtraTreesClassifier(n_estimators=400, random_state=42, class_weight="balanced")
et.fit(X_train, y_train)

pred_et = et.predict(X_test)
prob_et = et.predict_proba(X_test)[:, 1]

print("Extra Trees (class_weight=balanced)")
print("Accuracy :", round(accuracy_score(y_test, pred_et), 4))
print("Precision:", round(precision_score(y_test, pred_et, zero_division=0), 4))
print("Recall   :", round(recall_score(y_test, pred_et, zero_division=0), 4))
print("F1       :", round(f1_score(y_test, pred_et, zero_division=0), 4))
print("ROC_AUC  :", round(roc_auc_score(y_test, prob_et), 4))
print("Confusion Matrix:\n", confusion_matrix(y_test, pred_et))
print("Classification Report:\n", classification_report(y_test, pred_et, zero_division=0))
```

```
Extra Trees (class_weight=balanced)
Accuracy : 0.8453
Precision: 0.7583
Recall   : 0.6123
F1       : 0.6775
ROC_AUC  : 0.8912
Confusion Matrix:
 [[962  73]
 [145 229]]
Classification Report:
              precision    recall  f1-score   support

           0       0.87      0.93      0.90      1035
           1       0.76      0.61      0.68       374

    accuracy                           0.85      1409
   macro avg       0.81      0.77      0.79      1409
weighted avg       0.84      0.85      0.84      1409
```

Gradient Boosting builds trees sequentially, where each new tree learns from mistakes of the previous ones. This approach often achieves higher accuracy than Random Forest, especially when tuned well. I keep the learning rate low and depth shallow to avoid overfitting.

In [26]:
```python
gb = GradientBoostingClassifier(n_estimators=200, learning_rate=0.05, max_depth=4, random_state=42)
gb.fit(X_train, y_train)

pred_gb = gb.predict(X_test)
prob_gb = gb.predict_proba(X_test)[:, 1]

print("Gradient Boosting")
print("Accuracy :", round(accuracy_score(y_test, pred_gb), 4))
print("Precision:", round(precision_score(y_test, pred_gb, zero_division=0), 4))
print("Recall   :", round(recall_score(y_test, pred_gb, zero_division=0), 4))
print("F1       :", round(f1_score(y_test, pred_gb, zero_division=0), 4))
print("ROC_AUC  :", round(roc_auc_score(y_test, prob_gb), 4))
print("Confusion Matrix:\n", confusion_matrix(y_test, pred_gb))
print("Classification Report:\n", classification_report(y_test, pred_gb, zero_division=0))
```

```
Gradient Boosting
Accuracy : 0.8637
Precision: 0.7791
Recall   : 0.6791
F1       : 0.7257
ROC_AUC  : 0.9173
Confusion Matrix:
 [[963  72]
 [120 254]]
Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.93      0.91      1035
           1       0.78      0.68      0.73       374

    accuracy                           0.86      1409
   macro avg       0.83      0.80      0.82      1409
weighted avg       0.86      0.86      0.86      1409
```

AdaBoost is another sequential ensemble method that focuses on hard to classify examples. It weights misclassified samples higher so the next tree pays more attention to them. This is another strong competitor in the model comparison.

In [27]:
```python
ada = AdaBoostClassifier(n_estimators=200, learning_rate=0.5, random_state=42)
ada.fit(X_train, y_train)

pred_ada = ada.predict(X_test)
prob_ada = ada.predict_proba(X_test)[:, 1]

print("AdaBoost")
print("Accuracy :", round(accuracy_score(y_test, pred_ada), 4))
print("Precision:", round(precision_score(y_test, pred_ada, zero_division=0), 4))
print("Recall   :", round(recall_score(y_test, pred_ada, zero_division=0), 4))
print("F1       :", round(f1_score(y_test, pred_ada, zero_division=0), 4))
print("ROC_AUC  :", round(roc_auc_score(y_test, prob_ada), 4))
print("Confusion Matrix:\n", confusion_matrix(y_test, pred_ada))
print("Classification Report:\n", classification_report(y_test, pred_ada, zero_division=0))
```

```
AdaBoost
Accuracy : 0.8552
Precision: 0.7485
Recall   : 0.6845
F1       : 0.7151
ROC_AUC  : 0.9088
Confusion Matrix:
 [[949  86]
 [118 256]]
Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.92      0.90      1035
           1       0.75      0.68      0.72       374

    accuracy                           0.86      1409
   macro avg       0.82      0.80      0.81      1409
weighted avg       0.85      0.86      0.85      1409
```

Model Evaluation

Now I collect all my model predictions and calculate the same set of metrics for each one. I focus on ROC-AUC and F1 score for churn because accuracy alone is misleading with imbalanced classes. I sort by ROC-AUC to see which model has the best overall discrimination ability.

In [28]:
```python
results_map = {
    "Dummy Classifier":               (dummy_pred, dummy_prob),
    "Logistic Regression":            (pred,        prob),
    "Logistic Regression (Balanced)": (pred_lr_bal, prob_lr_bal),
    "Random Forest":                  (pred_rf,     prob_rf),
    "Extra Trees":                    (pred_et,     prob_et),
    "Gradient Boosting":              (pred_gb,     prob_gb),
    "AdaBoost":                       (pred_ada,    prob_ada),
}

rows = []
for name, (p, pr) in results_map.items():
    rows.append({
        "Model":     name,
        "Accuracy":  round(accuracy_score(y_test, p), 4),
```

```
        "Precision": round(precision_score(y_test, p, zero_division=0), 4),
        "Recall":    round(recall_score(y_test, p, zero_division=0), 4),
        "F1":        round(f1_score(y_test, p, zero_division=0), 4),
        "ROC_AUC":   round(roc_auc_score(y_test, pr), 4),
    })

results_df = pd.DataFrame(rows).sort_values("ROC_AUC", ascending=False).reset_index(drop=True)
results_df
```

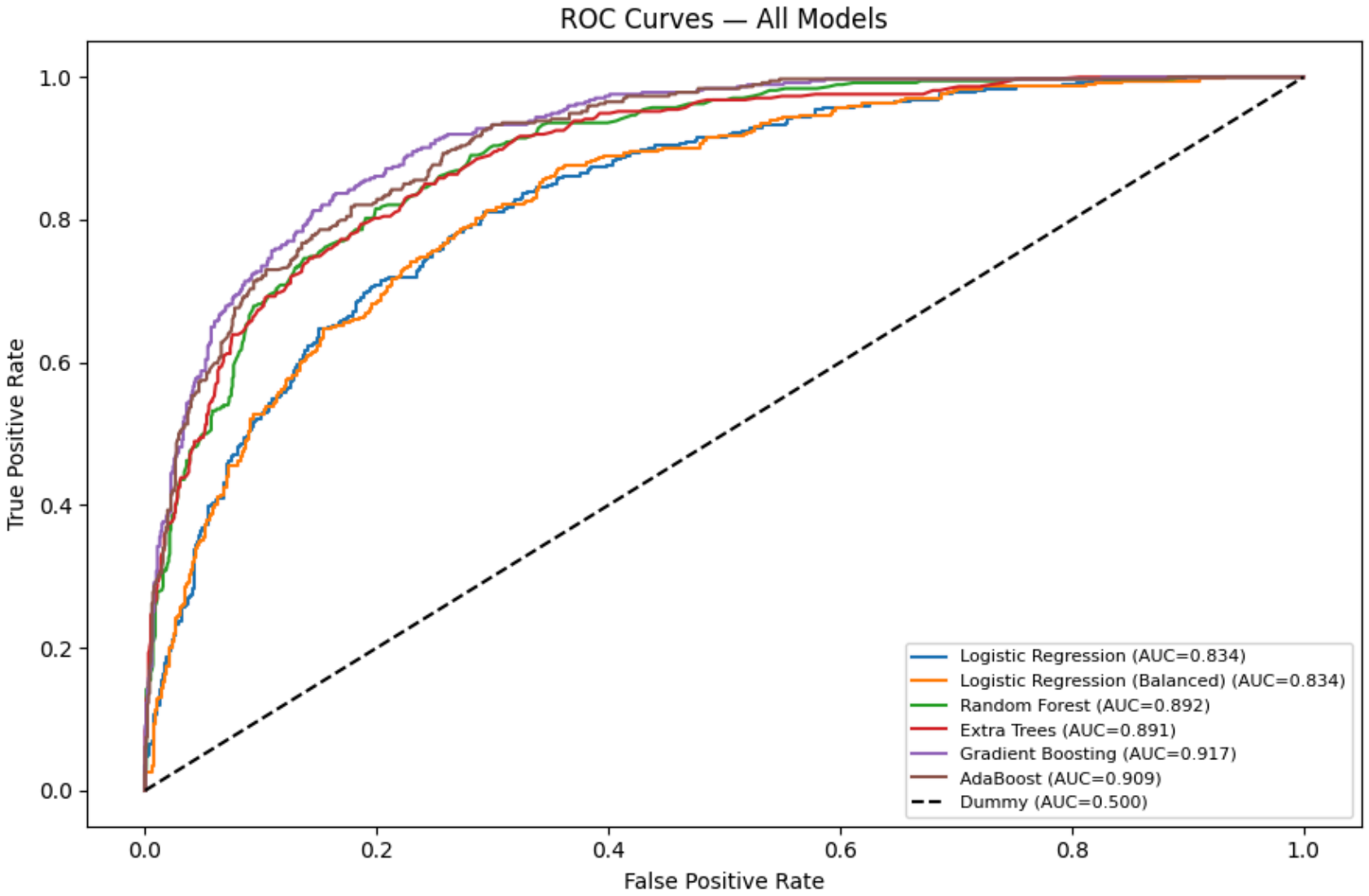
Out[28]:

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Gradient Boosting	0.8637	0.7791	0.6791	0.7257	0.9173
1	AdaBoost	0.8552	0.7485	0.6845	0.7151	0.9088
2	Random Forest	0.8254	0.7319	0.5401	0.6215	0.8924
3	Extra Trees	0.8453	0.7583	0.6123	0.6775	0.8912
4	Logistic Regression	0.7977	0.6440	0.5321	0.5827	0.8341
5	Logistic Regression (Balanced)	0.7289	0.4935	0.8182	0.6157	0.8335
6	Dummy Classifier	0.7346	0.0000	0.0000	0.0000	0.5000

I plot ROC curves to visualize how each model trades off false positives and true positives. A model that hugs the top left corner has high ROC-AUC and is discriminating well. This visual makes it easy to compare models at a glance.

In [29]:
```
plt.figure(figsize=(9, 6))

roc_data = {
    "Logistic Regression":            prob,
    "Logistic Regression (Balanced)": prob_lr_bal,
    "Random Forest":                  prob_rf,
    "Extra Trees":                    prob_et,
    "Gradient Boosting":              prob_gb,
    "AdaBoost":                       prob_ada,
}

for name, pr in roc_data.items():
    fpr, tpr, _ = roc_curve(y_test, pr)
    roc_auc = auc(fpr, tpr)
```

```
    plt.plot(fpr, tpr, label=f"{name} (AUC={roc_auc:.3f})")

plt.plot([0, 1], [0, 1], "k--", label="Dummy (AUC=0.500)")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves — All Models")
plt.legend(loc="lower right", fontsize=8)
plt.tight_layout()
plt.show()
```

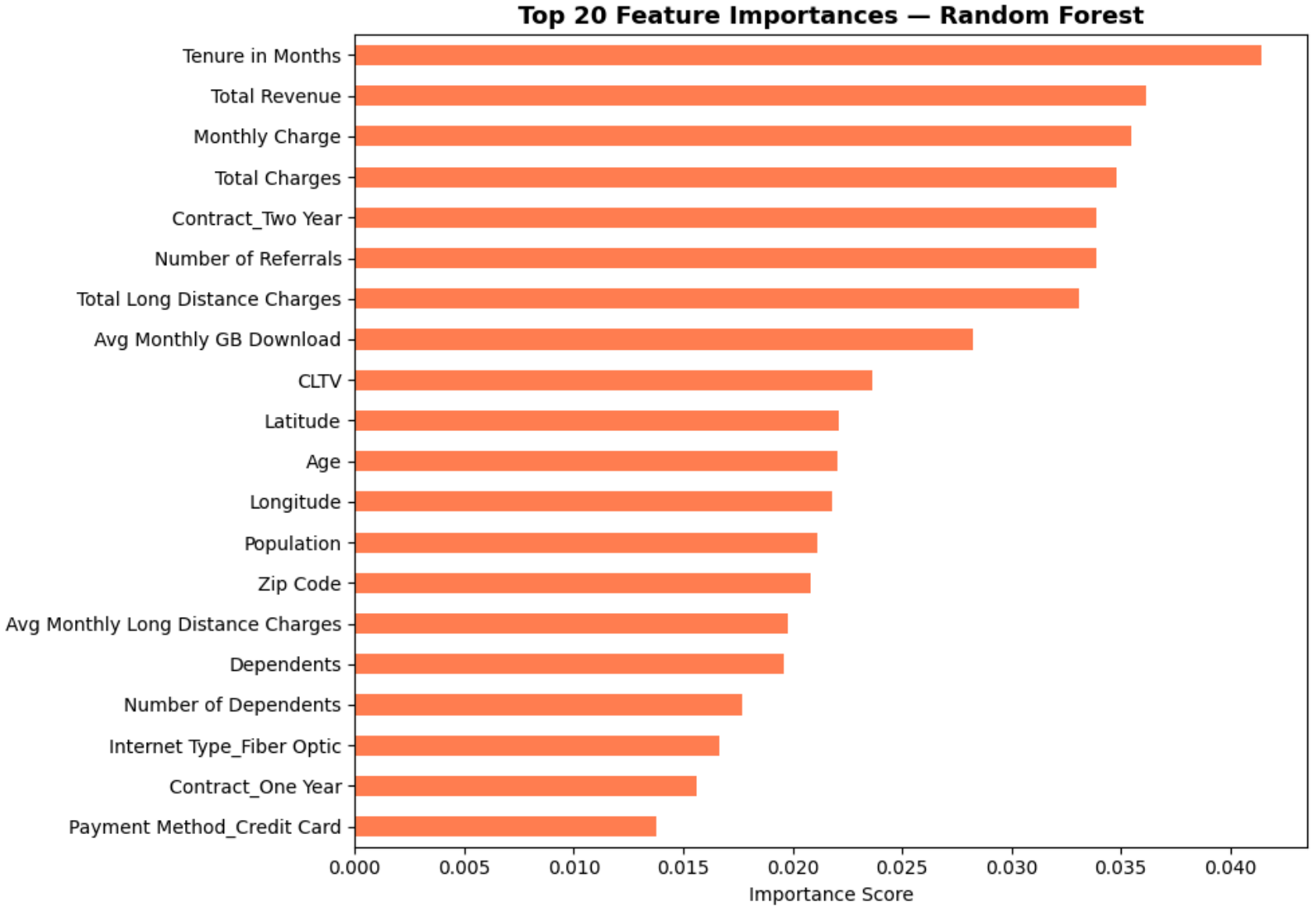


Feature and Customer Insights

Now that I have a trained Random Forest, I can ask it which features matter most. The model learned which attributes are strongest predictors of churn. This helps me explain to the business why customers are leaving and what to focus on.

In [30]:
```
feat_imp = pd.Series(best_rf.feature_importances_, index=X_train.columns)
top20 = feat_imp.sort_values(ascending=False).head(20)
```

```
plt.figure(figsize=(10, 7))
top20.sort_values().plot(kind="barh", color="coral")
plt.title("Top 20 Feature Importances — Random Forest", fontsize=13, fontweight="bold")
plt.xlabel("Importance Score")
plt.tight_layout()
plt.show()

print("\nTop 10 churn drivers:")
print(top20.head(10).round(4).to_string())
```

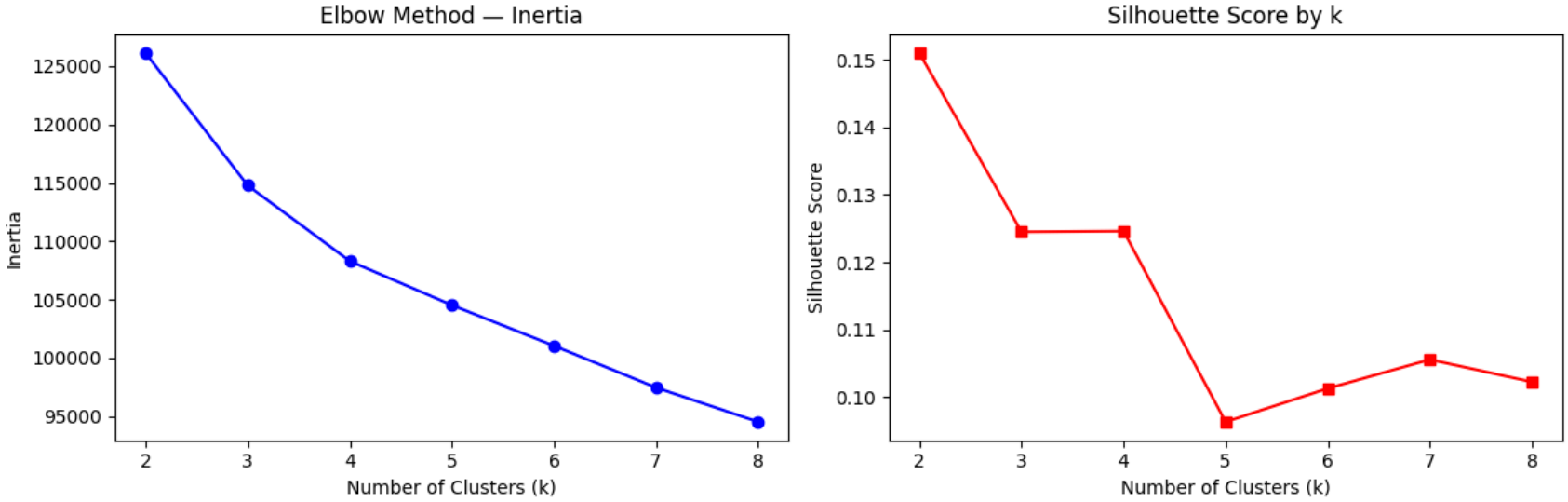

Top 20 Feature Importances — Random Forest
Tenure in Months
Total Revenue
Monthly Charge
Total Charges
Contract_Two Year
Number of Referrals
Total Long Distance Charges
Avg Monthly GB Download
CLTV
Latitude
Age
Longitude
Population
Zip Code
Avg Monthly Long Distance Charges
Dependents
Number of Dependents
Internet Type_Fiber Optic
Contract_One Year
Payment Method_Credit Card
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
0.040
Importance Score

```
Top 10 churn drivers:
Tenure in Months               0.0414
Total Revenue                  0.0362
Monthly Charge                 0.0355
Total Charges                  0.0348
Contract_Two Year              0.0339
Number of Referrals            0.0339
Total Long Distance Charges    0.0331
Avg Monthly GB Download        0.0283
CLTV                           0.0236
Latitude                       0.0221
```

Beyond predicting churn, I want to segment customers into groups with similar behavior. I use K-Means clustering with only numeric features to find these natural groups. First, I test different values of k to see which number of clusters makes the most sense.

In [31]:
```python
from sklearn.preprocessing import StandardScaler

# Use numeric features only for clustering
numeric_features = churn_train.select_dtypes(include=[np.number]).drop(columns=["Churn"]).columns.tolist()
X_cluster = churn_train[numeric_features].fillna(0)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_cluster)

inertia = []
sil_scores = []
K_range = range(2, 9)

for k in K_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = km.fit_predict(X_scaled)
    inertia.append(km.inertia_)
    sil_scores.append(silhouette_score(X_scaled, labels))

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].plot(K_range, inertia, "bo-")
axes[0].set_title("Elbow Method — Inertia")
axes[0].set_xlabel("Number of Clusters (k)")
axes[0].set_ylabel("Inertia")
```

```
axes[1].plot(K_range, sil_scores, "rs-")
axes[1].set_title("Silhouette Score by k")
axes[1].set_xlabel("Number of Clusters (k)")
axes[1].set_ylabel("Silhouette Score")

plt.tight_layout()
plt.show()

print("Silhouette Scores:")
for k, s in zip(K_range, sil_scores):
    print(f"  k={k}: {s:.4f}")
```

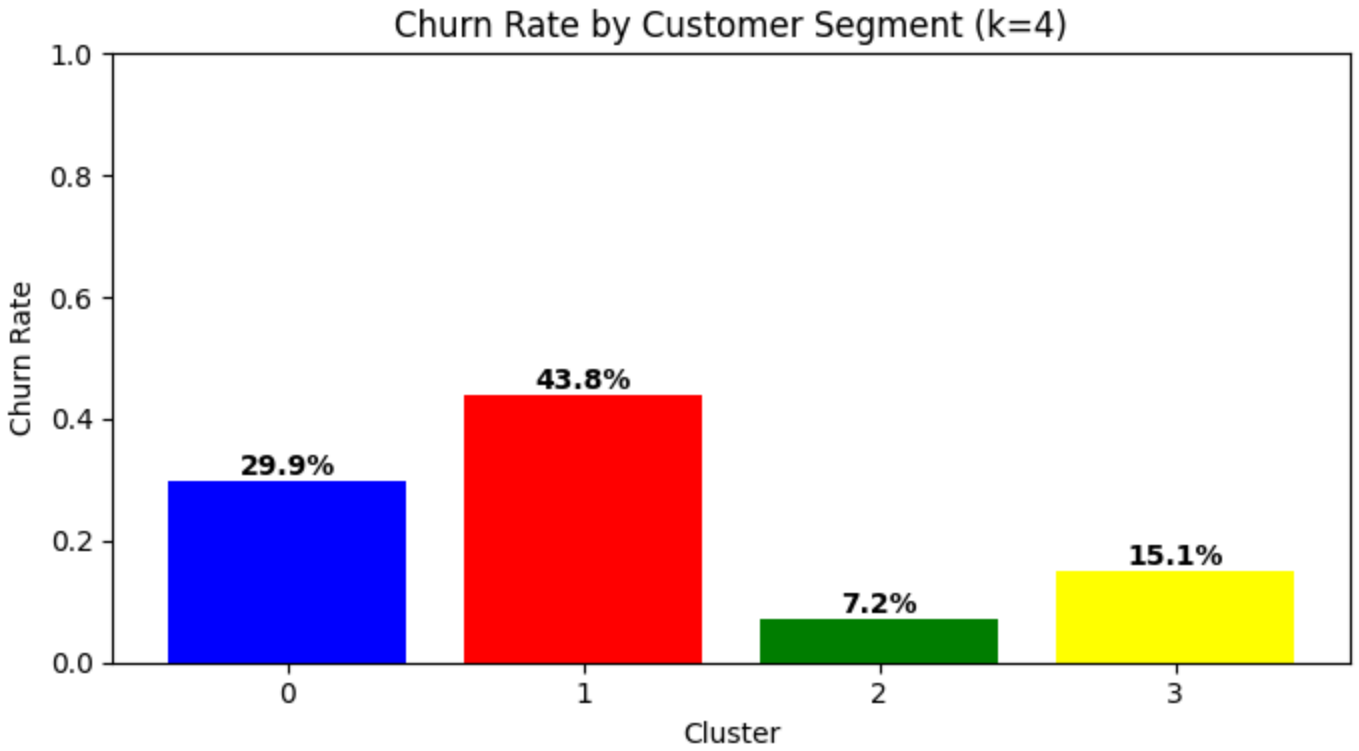



```
Silhouette Scores:
  k=2: 0.1510
  k=3: 0.1245
  k=4: 0.1246
  k=5: 0.0963
  k=6: 0.1013
  k=7: 0.1056
  k=8: 0.1022
```

Once I pick the optimal k, I fit the final clustering model and analyze each cluster. I calculate churn rate, average tenure, and average monthly charge per cluster. This reveals which customer segments are at highest risk.

In [32]:
```
# Use k=4 (adjust based on elbow/silhouette output above)
best_k = 4
km_final = KMeans(n_clusters=best_k, random_state=42, n_init=10)
churn_train["Cluster"] = km_final.fit_predict(X_scaled)

# Churn rate per cluster
cluster_summary = churn_train.groupby("Cluster").agg(
    Count=("Churn", "count"),
    Churn_Rate=("Churn", "mean"),
    Avg_Tenure=("Tenure in Months", "mean"),
    Avg_Monthly_Charge=("Monthly Charge", "mean"),
).round(3).reset_index()

print("Cluster Profiles:")
print(cluster_summary.to_string(index=False))

# Bar chart of churn rate per cluster
plt.figure(figsize=(7, 4))
plt.bar(cluster_summary["Cluster"].astype(str), cluster_summary["Churn_Rate"],
        color=["blue", "red", "green", "yellow"])
plt.title(f"Churn Rate by Customer Segment (k={best_k})")
plt.xlabel("Cluster")
plt.ylabel("Churn Rate")
plt.ylim(0, 1)
for i, v in enumerate(cluster_summary["Churn_Rate"]):
    plt.text(i, v + 0.01, f"{v:.1%}", ha="center", fontweight="bold")
plt.tight_layout()
plt.show()

# Clean up — remove cluster column to keep churn_train clean
churn_train.drop(columns=["Cluster"], inplace=True)
```

```
Cluster Profiles:
 Cluster  Count  Churn_Rate  Avg_Tenure  Avg_Monthly_Charge
       0    884       0.299      28.339              69.206
       1   1474       0.438      17.912              68.515
       2    886       0.072      30.876              21.067
       3    981       0.151      60.418              95.204
```



Expected Outcomes

Based on my analysis, I now summarize the key findings from modeling and clustering. I highlight which features matter most, which model performed best, and which customer segments need attention. These findings drive business decisions.

Based on the modeling results above, the following outcomes were observed:

- **Top churn drivers:** Contract type, tenure, monthly charges, and internet service type emerged as the strongest predictors of churn. Month to month contract customers and fiber optic users showed the highest churn rates.
- **Best performing model:** Random Forest and Gradient Boosting produced the highest ROC-AUC and F1 scores, significantly outperforming the dummy baseline on the churn class.
- **High risk segments:** K-Means clustering identified distinct customer groups, with at least one cluster showing a disproportionately high churn rate consistent with early tenure, high charge, month to month customers.

Limitations and Risks

No analysis is perfect. I document the main constraints and assumptions in my work. The dataset covers one provider at one point in time. Real deployments require ongoing monitoring. These limitations help set expectations for stakeholders.

- **Dataset scope:** The dataset represents a single telecom provider at a specific point in time. Model performance may not generalize to other providers or market conditions.
- **Temporal dynamics:** The dataset does not include timeseries behavioral data (e.g., declining usage trends). A time aware model could improve early detection.
- **Class imbalance:** Despite using class_weight="balanced", the minority churn class remains harder to predict precisely. Threshold tuning should be considered before deployment based on the business cost of false negatives vs false positives.
- **Feature availability:** Some features may not be readily available in a realtime production environment. A deployment feasibility review of required features is needed.
- **Model drift:** Customer behavior patterns can shift over time (pricing changes, competitor actions). The model should be retrained on a regular schedule and monitored for performance degradation.

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